

Section: 3.2 – MECHANICALLY FASTENED JOINTS**Chapter: 3260****Title: NON-STRUCTURAL SHIMS****Revision: 0.4****Date: 22/03/2022****1 INTRODUCTION**

The standard methods and allowable values available for the strength check of riveted joints will be applied to joints with non-structural shims with thickness up to the default limits authorized typically in part drawings within Airbus DS (Ref. 8):

Type of shim	Metallic structures	Composite or hybrid structures
Liquid	0.2 mm	0.5 mm
Solid	0.8 mm	1.2 mm

That approach is supported by past experience and tests (it is also recommended that shim thickness does not exceed 0.25 times diameter ratio to avoid fatigue issues in metal plates, see Ref. 2).

The following paragraphs present the corrections that must be applied in case the non-structural shimming exceeds those limitations.

Limitations

- ✓ This revision of the chapter has been released in advance to any discussion and agreement within Airbus DS division, in order to cover the need of a legacy project currently running and managed by Airbus DS - Spain site. Thus, this methodology is applicable only to legacy projects under Airbus DS - Spain perimeter, and NOT valid for transnational Airbus DS projects.
- ✓ The method presented herein is extracted from what was proposed to certify the structure of the Fan Cowl of the A320Neo (version powered with the CFM LEAP-X engine), produced by former Airbus Military division. Full document reference is shown as Ref. 1 in list of references.

Method for computing shim effect in bolt/rivet effect is also a standard practice in support to production quality issues and in-service repairs within Airbus DS – Spain (see Ref. 5).

✓ Potential impact in bearing strength of the attached plates is not discussed in this revision of the chapter

2 METHOD DESCRIPTION

2.1 SHIM IMPACT IN THE STIFFNESS OF THE JOINT

SHEAR STIFFNESS

Shear stiffness will be still calculated with to Huth's assuming no significant impact is introduced by the presence of the non-structural shim.

Fastener flexibility (inverse of stiffness) is calculated according to Huth's proposal as:

$$C = \left(\frac{t_1 + t_2}{2d} \right)^a \frac{b}{n} \left(\frac{1}{t_1 E_1} + \frac{1}{n t_2 E_2} + \frac{1}{2 t_1 E_3} + \frac{1}{2 n t_2 E_3} \right)$$

Group I—bolted metallic joints	$a = 2/3; b = 3.0$
Group II—riveted metallic joints	$a = 2/5; b = 2.2$
Group III—bolted graphite/epoxy joints	$a = 2/3; b = 4.2$

Figure 1. Huth's formula for fastener flexibility (see Ref. 4, original, and translation Ref. 4)

Where t_1 and t_2 are the thicknesses of the plates, d the rivet (shank) diameter, E_1 , E_2 are the Young moduli of plates (1,2) and E_3 the Young modulus of the bolt/rivet. Values of parameters a and b are stated in previous figure. Parameter n is equal to 1 in single shear joints and equal to 2 in double shear joints. Notice plate 1 is the central plate in double shear configurations.

The fastener stiffness is the inverse of the flexibility, thus:

$$K = 1 / C$$

TENSION STIFFNESS

In this case, the shim thickness is added to the term computing the flexibility of the bolt grip when rivet/bolt is loaded in tension. Using previous notation :

$$\text{Flexibility of rivet} \quad C_{\text{rivet}} = L / F = L / (EA)_{\text{rivet}} = (t_1 + t_2 + t_{\text{shim}}) / (E_3 \cdot \pi \cdot d^2 / 4)$$

...where d is the rivet diameter.

2.2 SHIM IMPACT IN THE STRENGTH OF THE JOINT

BOLT STRENGTH

The shim will be accounted in the strength check of the bolt by adding a bending moment to the internal loads levels of the shank cross section, as the shear load is transferred with an offset between the plates equal to the shim thickness:

- ✓ Shear failure index: $R_s = Q / Q_{all}$
- ✓ Tension failure index: $R_t = T / T_{all}$
- ✓ Bending failure index: $R_b = \Delta M / M_{all} = (Q \cdot t_{shim} / 2) / M_{all}$

Note: This approach does not mean that bolt-beam ends and supports are assumed to be at shim faces. Indeed, the consideration made is that the thickness offset is accounted in the nominal design shear allowable of the bolt.

Reserve Factor: $RF = 1 / [R_b + (R_t^2 + R_s^2)^{0.5}]$

Note: R_b correction has been added, not only to R_t term, but to the resultant failure index (as R_s term implicitly accounts for bending as noted above), in order to get a straight forward expression of the Reserve Factor.

The allowable bending moment of the shank section (M_{all}) can be calculated by means of FLEXA software to account for plastic bending in ultimate loads.

No correction shall be applied to the strength analysis of the nut/collar (vs tension load in the bolt).

One assumption beneath this last approach is that the extra bending introduced by the shear transfer with offset is reacted mainly in differential bearing stress through the thickness (some differential bearing is already present in any lap-shear joint, even without shim, and accounted in design allowable values).

METALLIC PLATES

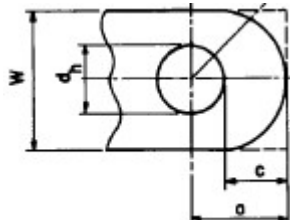
Left for further work.

No correction shall be applied to typical strength analyses (bearing, base bending,...) even if non-structural shims are present within the plates.

Regarding bearing failure in metallic plates, for which the traditional no-correction approach may not appear intuitive, classical bibliography shows its consistency for typical designs of riveted joints. Either Bruhn D1.11 (Ref. 7) or ESDU91008 (Ref. 6), propose shear-bearing failure curves of lugs proportional to the bearing area ($d \cdot t$, i.e. hole diameter times the thickness), to the ultimate stress of the material, and to a factor function of the edge distance and the lug thickness. For typical designs of riveted joints (lug in L-LT material plane, ratio e/d at least 2.2 and ratio t/d greater than 0.3), that factor

is equal to 2, therefore the bearing ultimate stress becomes $2 \cdot F_{tu}$, which is consistent with the F_{bru} values given for metal alloys.

Notice this method is applicable no matter the value of the shim thickness (gap in ESDU or Bruhn).



The load to cause failure due to shear-bearing is given by the expression:

$$P_{gux} = K_{gux} f_{tum} d_h t$$

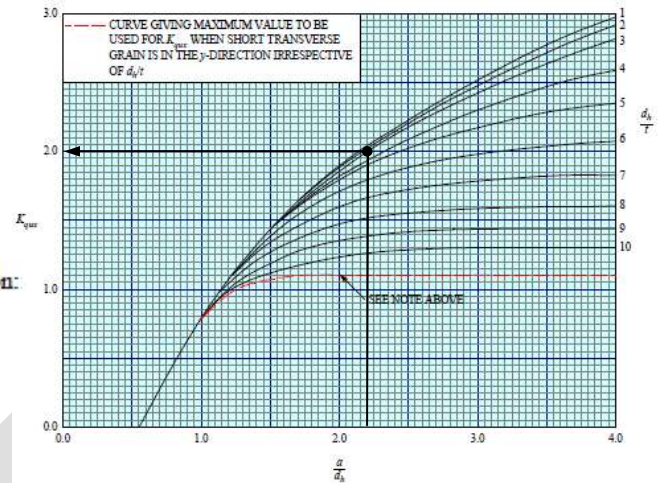


Figure 2. Extracts from ESDU 91008 (Ref. 6)

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
1	-	As subscript, refers to plate 1 (central plate in double shear joint)
2	-	As subscript, refers to plate 2
3	-	As subscript, refers to the rivet/bolt
a, b	-	Parameters of the Huth's formula
A	mm ²	Area of the cross-section
C	mm/N	Fastener flexibility
d	mm	Diameter of the fastener shank
E	MPa	Young modulus
F	N	Force
FEM	-	Finite Elements Model
K	N/mm	Fastener stiffness
L	mm	Length/grip of the fastener
n	-	Parameter of the Huth's formula, equal to 2 for configuration sin double shear joints, and equal to 1 in single lap shear
N/A	-	Not Applicable
t	mm	Thickness of the plate

w mm Effective width

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	ME-DN-NT-120004	Is. E	STRESS ANALYSIS METHODS FOR SIZING STRUCTURAL PARTS OF FAN COWL A320NEO
Ref. 2	NT-T-ADS-92042	Is. B	PROCEDIMIENTO DE CALCULO DE UNIONES REMACHADAS METALICAS
Ref. 3	[H. Huth]	1984	Zum Einfluß der Nietnachgiebigkeit mehrreihiger Nietverbindungen auf die Lastübertragungs und Lebensdauervorhersage
Ref. 4	[H. Huth]	1986	Influence of Fastener Flexibility on the Prediction of Load Transfer and Fatigue Life for Multiple-Row Joints
Ref. 5	TAE-0-NT-140002	Is. B	Static, Fatigue and Damage Tolerance Methodologies for Justification of Structural Repairs
Ref. 6	ESDU 91008	1991	STRENGTH OF LUGS
Ref. 7	[E.F. Bruhn]	1973	Analysis and Design of Flight Vehicle Structures
Ref. 8	[Airbus] I+D-P-061	Rev. C	Suplementos líquidos para ajuste

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
FLEXA	3.9	Strength check of the section of a straight beam made of an elasto-plastic material and submitted to tension/compression axial loads and bending

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.4 22/03/2022	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEPAP-TL4)</i>	G. Baños 22/03/2022	Name DD/MM/2022

Table 3. *Approval signatures table for last revision of this chapter*